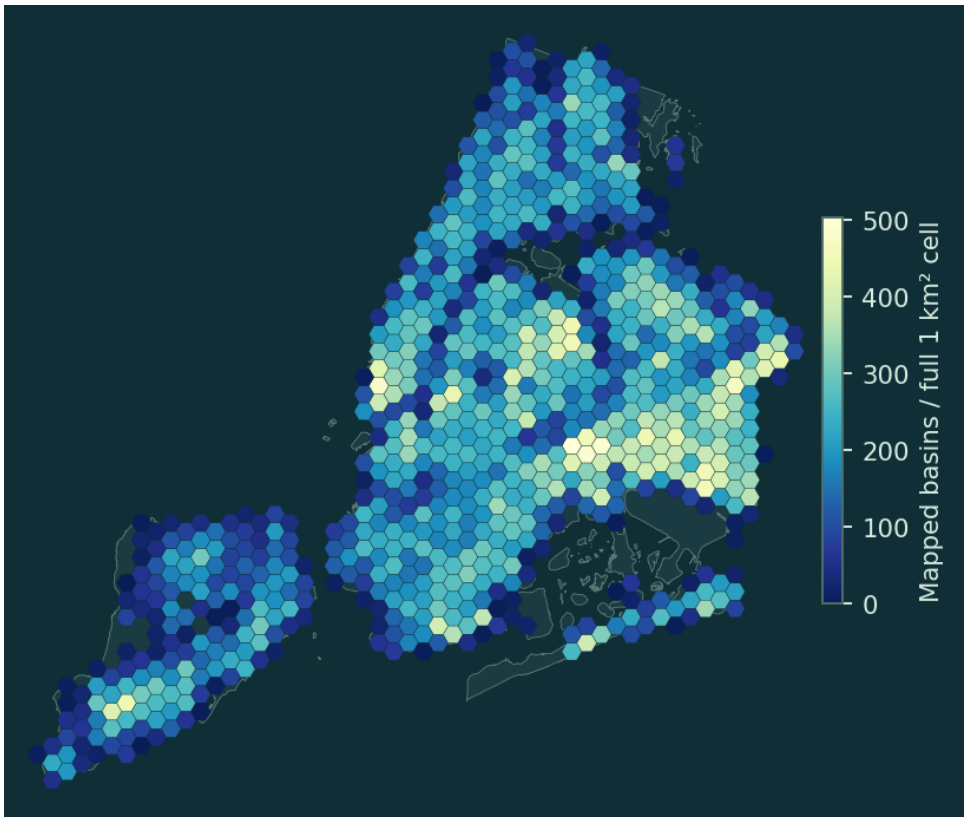
The Iron Grid

Mapping NYC's hidden stormwater network

An interactive comparison of catch basin infrastructure, reported problems and historical precipitation.



Why this matters: Making the drainage network and public reports visible together can support better questions for field investigation.

Live prototype: https://iron-grid-nyc.wrylake10.chatgpt.site

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Theme: Undercurrents | Extended due date: October 28, 2026

Draft: update the final host link after GitHub Pages publication and add observations from your own browser walkthrough.

From a location map to a useful question.

Primary question

Where do mapped catch basins and reported catch basin problems overlap, and how do report patterns change around wet periods?

Audience and use case

Civil engineers and environmental planners can identify locations to examine more closely before a site visit. Community boards and residents can inspect the same public evidence. The prototype supports exploratory screening; hydraulic design and flood prediction require additional measurements.

Connection to Undercurrents

The project reveals two undercurrents: the physical drainage network below the street, and the less visible stream of public reports that draws attention to it. Rainfall adds an environmental cycle that gives those reports temporal context.

What changed after the proposal

Proposal foundation	Part 2 addition
Inventory coordinates and basin IDs	Report types, recovered creation dates, official borough polygons, Central Park precipitation
Citywide cells to asset points	Linked borough/type/date filters, brushing, inspection, two-cell comparison and CSV export
Questions about density, old streams and equity	These remain hypotheses. The current tool shows observed locations and reports without asserting causal explanations.

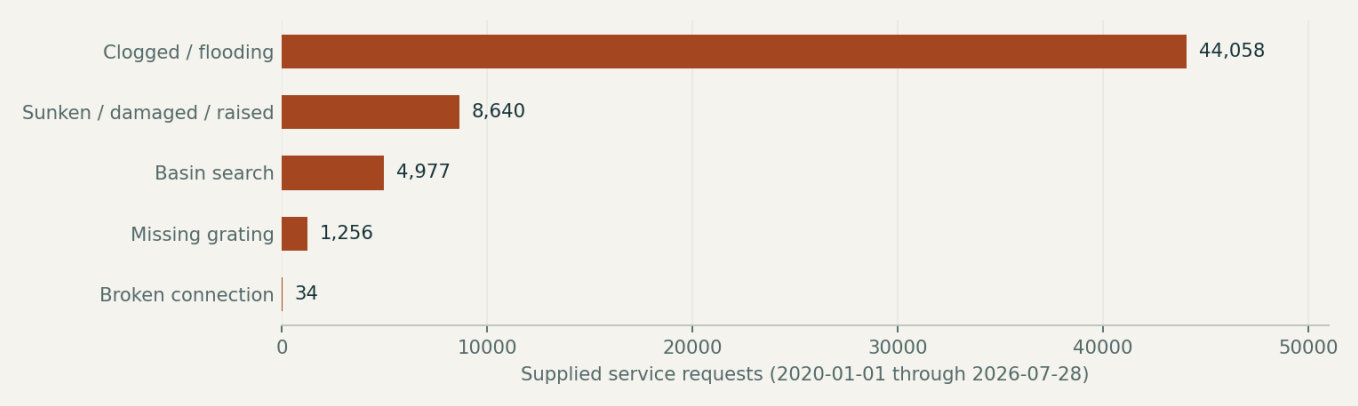
Gallery criteria explicitly addressed

Understanding: readable aggregation and record inspection. **Usefulness:** a way to select a place or period for further investigation. **Design:** progressive disclosure and a numeric alternative to the map. **Transparency:** downloadable audit/manifest, source links and preparation code. [6]

GitHub Pages publication is the final hosting handoff because the assignment names GitHub Pages or Observable. The current demo is publicly hosted on Sites.

EDA: inspect the records before the pattern.

Input	Rows	Fields	Nulls / duplicate IDs
Catch basin inventory [1]	154,212	3 retained	0 / 0
Supplied 311 reports [2]	58,965	4 original	0 / 0
Central Park PRCP [4]	2,401 days	Day, mm, flags	0 unavailable days



Observed: 44,058 supplied reports (74.7%) concern clogged/flooding basins. This establishes a promising category for exploration; it does not establish how many distinct basins failed.

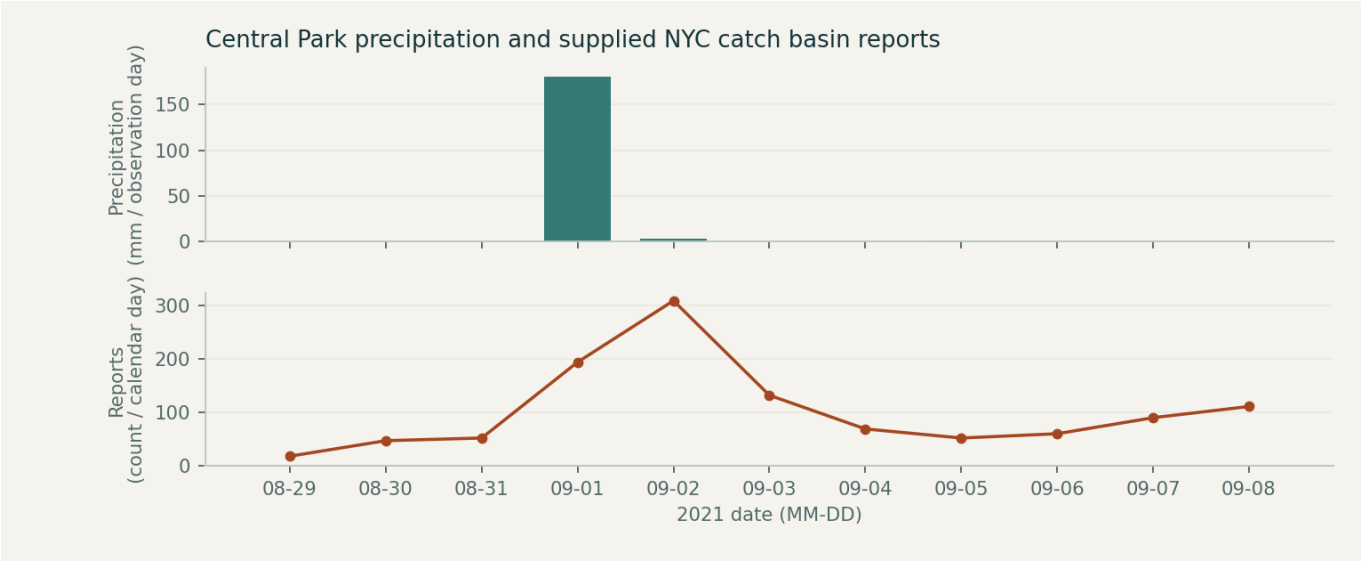
Cleaning and preservation

The supplied basin file has three retained fields; the earlier slide described a six-column source export. Both inputs have unique IDs and valid coordinates in a broad NYC envelope. There are 154,196 distinct basin coordinates and 36,992 distinct report coordinates. Different IDs at the same location remain separate records.

All 58,965 report creation dates were restored by exact request key. Dates span January 1, 2020 to July 28, 2026. The complaint column labeled UNITID is renamed REQUEST_ID in the enriched export; it is never merged with asset UNITID.

Full-resolution borough polygons [3] assign records spatially. The 29 unassigned basins and 8 unassigned reports remain in the citywide view. Original hashes, preparation code and an audit are provided.

Rain creates context, not a causal result.



Actual data around September 1, 2021, with separate units in each panel. Central Park recorded 181.1 mm on the observation day labeled September 1; the supplied file has 309 requests dated September 2. These adjacent patterns motivate inspection, not a causal estimate. [2,4]

A meaningful temporal interaction

Brushing changes the report period used in the map, breakdown, rankings and cell comparison. The full timeline stays visible so visitors keep context. Date inputs provide the same selection through a keyboard.

A carefully bounded comparison

The station measures one place, not rainfall in every borough. Daily totals do not measure the short bursts of rainfall that can exceed drainage capacity. Observation-day boundaries may differ from 311 calendar days, and reports can arrive late.

NOAA PRCP is converted from tenths of millimeters to millimeters. Nonblank quality flags and negative missing codes are excluded; trace flags remain annotated zeros. Missing observations would appear as gaps. Precipitation includes liquid equivalents of frozen precipitation. [4,5]

Question to take into the field

Do cells with repeated clogged/flooding requests around a wet period warrant closer inspection of inlet condition, street grade or downstream constraints? The map can identify a candidate cell; field and engineering data must answer the question.

Onboarding: enter, orient, then explore.

An optional three-step dialog is available from the landing area and the persistent How to use control. Visitors can move back, close it or enter the explorer.

01 / A city under the city

Catch basins collect street runoff and connect it to the sewer network. This map begins with 154,212 mapped entrances to that hidden system.
Intent: connect the dataset to a familiar object before introducing controls.

02 / Turn a pattern into a question

Choose a borough and switch the map to report counts. Drag across the rainfall timeline, or use the date inputs, to explore a period.
Intent: introduce one exploration sequence, then cell details and comparison.

03 / Keep the limits in view

A report is a service request, not proof of a failed drain. Basin locations do not reveal drainage capacity; Central Park rain is one local observation.
Intent: establish responsible interpretation before drawing conclusions.

Experience design

Orientation: familiar drains, a clear question and a named audience. **Invitation:** choose a borough and inspect a cell. **Tempo:** counts, then points, then records. **Framing:** persistent dates, legends and layer meanings. **Accessibility:** visible focus, keyboard controls, numeric table, reduced-motion support and date inputs.

These are scripted onboarding frames. The figures in this PDF are data visualizations and storyboard copy, not unverified browser screenshots or claims of completed user testing.

Methods, ethics and known limits.

Choice	Reason and limitation
Spatial aggregation	Regular 1 km² hexagons in a local equal-area projection. Edge cells have varying land fractions; counts are not normalized by roads or population.
Reports per 100 basins	Selected reports / fixed inventory count × 100 in a cell. A spatial ratio, not a failure rate. No denominator means unavailable; small denominators are flagged.
311 representation	Requests may repeat and depend on access, awareness and delays. Low reporting does not prove adequate drainage.
Temporal coverage	Fixed supplied inventory; report coverage ends July 28, 2026. Original export date and cleaning history were not supplied. Partial years are not full-year totals.
Privacy and consent	Public asset/request IDs, coordinates, categories and dates only. No complainant identities, addresses or contacts. Public availability is not proof of consent.
Engineering scope	No elevation, capacity, pipe network, catchment area, installation history or maintenance outcome is inferred. Request closure would not establish a repair.

Reproducibility

Python profiles the inputs, restores dates by unique_key, assigns boroughs with full geometry, computes hex membership and parses NOAA flags. Boundaries are simplified only for display. The browser loads a static prepared snapshot rather than live API queries.

Validation performed

Checks passed for totals, borough/type/date filters, ID separation, fixed denominators, missing-denominator ratios, daily counts, hex area/membership and required assets. Browser interaction and visual QA were unavailable. A complete browser walkthrough remains necessary.

Reflection and an achievable finish.

What works

Two scales make the inventory interpretable: citywide cells and individual records. Linked periods and report types create an explorable question. Recovering the dates adds time without replacing the supplied cohort.

What is unresolved

The map has not been user-tested. Counts depend on land area, street layout and reporting behavior. One station cannot describe local storm variability. The report cohort is not proven complete, and the fixed inventory may not match historical conditions.

Where it goes next

Complete browser testing and GitHub Pages publication first. Next, add a labeled NYC stormwater flood scenario [7]. A terrain layer needs a documented DEM resolution, date, projection and vertical datum. Multiple gauges or a documented radar product would improve rainfall context. These are proposed extensions.

Rubric	Current evidence	Remaining action
Concept / 20%	Hidden network, report stream, audience	Explain the screening use case
EDA & ethics / 20%	Profiles, figures, preparation and limits	Verify interpretations
Creativity / 15%	Grid-to-street reveal and time comparison	Refine after one user trial
Interactive / 25%	Zoom, brush, filters, inspection, export	Run browser acceptance checks
Feasibility / 10%	Buildless site; bounded next layers	Freeze features by Oct 26
Presentation / 10%	PDF, source bundle and live demo	Update final URL and rehearse

Add 2-3 specific observations from your own walkthrough to this reflection before handing in the final PDF.

Sources and final submission handoff.

[1] NYCDEP Citywide Catch Basins

Open the official source. Uploaded inventory; original export date unknown.

[2] 311 Service Requests from 2020 to Present

Open the official source. Uploaded report cohort; dates restored October 17, 2026.

[3] Borough Boundaries

Open the official source. Official geometry for spatial classification.

[4] NOAA GHCN-Daily, Central Park USW00094728

Open the official source. PRCP subset: January 1, 2020 to July 28, 2026.

[5] NOAA GHCN-Daily documentation

Open the official source. Units and flags; CSV layout in readme-by_station.txt.

[6] NYC Open Data Gallery criteria

Open the official source. Criteria and dataset-list requirement, reviewed October 17, 2026.

[7] NYC Stormwater Flood Maps

Open the official source. Proposed next layer; not in the current prototype.

Milestone handoff

Submit the public GitHub Pages link, revised PDF and source/preparation evidence. Demonstrate: onboarding, one filter, a brushed period, a record and a cell comparison.

D3.js 7.9.0 (ISC) is bundled locally. NOAA: Menne et al., GHCN-Daily Version 3, DOI 10.7289/V5D21VHZ; subset and access date above. Neither NYC nor NOAA endorses this project.

The City Gallery application is separate from the course submission. Its official rules and ownership statements require the creator review before an actual submission. No Gallery application has been sent.